CLINICAL STUDY REPORT

A Phase 1 Open-label, Dose-escalation Study of Kelvaparib in Adults with Advanced Solid Tumours

Study KLV-7702

Investigational product	Kelvaparib (KLV-3308), oral PARP1-selective inhibitor
Indication studied	Advanced solid tumours with a homologous recombination repair defect
Development phase	Phase 1
Sponsor	Marnholt Oncology GmbH
Protocol number	KLV-7702
First subject enrolled	31 October 2016
Data cut-off date	3 April 2019
Database lock	1 May 2019
Report date	16 October 2019
Report status	Final
Good Clinical Practice	Conducted in accordance with the Declaration of Helsinki and ICH Good Clinical Practice.

2 SYNOPSIS

Name of sponsor	Marnholt Oncology GmbH
Name of finished product	KLV-3308
Name of active ingredient	Kelvaparib (KLV-3308)
Title of study	A Phase 1 Open-label, Dose-escalation Study of Kelvaparib in Adults with Advanced Solid Tumours
Study number	KLV-7702
Phase of development	Phase 1
Studied period	31 October 2016 to 3 April 2019
Analysis	Final analysis
Objectives	Primary: maximum tolerated dose and recommended phase 2 dose. Secondary: duration of response, progression-free survival, overall survival and the incidence and severity of treatment-emergent adverse events, and pharmacokinetics.
Methodology	An open-label, multicentre, non-randomised, sequential dose-escalation study using a 3+3 design, followed by expansion at the dose selected as the recommended phase 2 dose. Approximately 58 subjects across 4 dose cohorts.
Number of subjects	58 subjects received at least one dose of kelvaparib (6 at 100 mg, 9 at 200 mg, 31 at 300 mg, 12 at 400 mg).
Diagnosis and main criteria for inclusion	Adults with histologically or cytologically confirmed advanced solid tumours with a homologous recombination repair defect, measurable disease assessable by RECIST version 1.1, ECOG performance status 0 to 2, adequate organ function.
Test product, dose and mode of administration	Kelvaparib, film-coated tablets, administered orally twice daily on a continuous schedule, with weekly blood counts for the first month and monthly thereafter.
Duration of treatment	Until disease progression, unacceptable toxicity or withdrawal of consent. Median duration 122 days.
Criteria for evaluation - efficacy	Best overall response by RECIST version 1.1, and duration of response, progression-free survival, overall survival.
Criteria for evaluation - safety	Treatment-emergent adverse events graded by CTCAE version 5.0, laboratory parameters and vital signs.
Statistical methods	No formal hypothesis testing is planned. Sample size follows the operating characteristics of the 3+3 design and the number of subjects required to characterise safety at the selected dose. Proportions are presented with exact Clopper-Pearson 95 per cent confidence intervals and time-to-event endpoints estimated by the Kaplan-Meier method.
Summary - efficacy	The overall response rate was 16 of 55 subjects (29.1 per cent, 95 per cent CI (17.6, 42.9)). Complete response in 4 subjects (7.3 per cent). Median progression-free survival 8.5 months.
Summary - safety	Grade 3 or higher events in 41 subjects (70.7 per cent), serious events in 22 (37.9 per cent). Nausea was the most frequent event (37 subjects, 63.8 per cent). Myelodysplastic syndrome or acute myeloid leukaemia, an event of special interest, occurred in 2 subjects (3.4 per cent). 10 discontinued because of an adverse event; 8 deaths were reported.

Conclusion	The maximum tolerated dose was 300 mg orally twice daily: dose-limiting toxicities occurred in 0 of 6, 2 of 7, 5 of 28 and 4 of 12 DLT-evaluable subjects at 100, 200, 300 and 400 mg, and 300 mg is the recommended phase 2 dose. Kelvaparib produced antitumor activity in advanced solid tumours with a homologous recombination repair defect (overall response rate 29.1 per cent); efficacy observations are exploratory.
Date of report	16 October 2019

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4 LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation	Definition
ALT	alanine aminotransferase
AST	aspartate aminotransferase
AUC0-12	area under the concentration-time curve over one dosing interval (0 to 12 hours)
AUC0-24	area under the concentration-time curve from 0 to 24 hours
BRCA	breast cancer susceptibility gene
BRCA1	breast cancer susceptibility gene 1
BRCA2	breast cancer susceptibility gene 2
CA-125	cancer antigen 125
CI	confidence interval
CL/F	apparent oral clearance
Cmax	maximum plasma concentration
CR	complete response
CTCAE	Common Terminology Criteria for Adverse Events
Ctrough	trough plasma concentration
CV	coefficient of variation
DLT	dose-limiting toxicity
ECOG	Eastern Cooperative Oncology Group
ICH	International Council for Harmonisation
KLV-3308	kelvaparib (compound code)
MedDRA	Medical Dictionary for Regulatory Activities
MTD	maximum tolerated dose
NE	not estimable
PFS	progression-free survival
PR	partial response
Q1	first quartile
Q3	third quartile
RECIST	Response Evaluation Criteria in Solid Tumours
RP2D	recommended phase 2 dose
SD	stable disease
t1/2	terminal elimination half-life
TEAE	treatment-emergent adverse event
tmax	time to maximum plasma concentration

5 ETHICS

5.1 Independent ethics committee or institutional review board

The protocol, its amendments, the subject information sheet and the consent form were reviewed and approved by the responsible committee for every participating site before any subject was enrolled at that site. Approvals are held in the trial master file.

5.2 Ethical conduct of the study

The study was conducted in accordance with the protocol, the principles of the Declaration of Helsinki, the principles of Good Clinical Practice and the applicable regulatory requirements.

5.3 Subject information and consent

Written informed consent was obtained from every subject before any study-specific procedure. Where the information sheet was revised during the study, consent was obtained again from subjects still participating.

6 INVESTIGATORS AND STUDY ADMINISTRATIVE STRUCTURE

The study was conducted at the sites listed in Appendix 16.1.4, identified by number. Investigator names are not recorded in this report. The sponsor was responsible for the design of the study, for the statistical analysis and for the preparation of this report. Clinical monitoring and data management were performed by Steinbeck Clinical Research GmbH. Laboratory analyses were performed at a central laboratory. Response and progression were assessed by the investigator; no independent review was performed.

7 INTRODUCTION

This report describes study KLV-7702, A Phase 1 Open-label, Dose-escalation Study of Kelvaparib in Adults with Advanced Solid Tumours. The study was conducted to evaluate Kelvaparib in subjects with advanced solid tumours with a homologous recombination repair defect. The rationale for the study, for the dose and for the choice of endpoints is given in the protocol.

8 STUDY OBJECTIVES

Table 8-1 Study objectives

Type	Objective
Primary	Maximum tolerated dose and recommended phase 2 dose
Safety	To characterise the safety and tolerability of Kelvaparib.

9 INVESTIGATIONAL PLAN

9.1 Overall study design and plan

An open-label, multicentre, non-randomised, sequential dose-escalation study using a 3+3 design, followed by expansion at the dose selected as the recommended phase 2 dose. Approximately 58 subjects across 4 dose cohorts.

9.2 Discussion of study design, including choice of control

The design was appropriate to the question. A single-arm design was used because the primary endpoint, dose-limiting toxicity in the first cycle, is observed within each subject and needs no concurrent control. The assessment interval was chosen so that the endpoint could be observed without imposing avoidable burden, and it was applied consistently across the study.

9.3 Selection of study population

Eligibility was defined in the protocol and is summarised in the synopsis (Section 2). Every subject was required to meet all inclusion criteria and none of the exclusion criteria; no waivers were granted.

9.4 Treatments

Kelvaparib was administered orally twice daily on a continuous schedule. Dose modification followed the rules in the protocol. Study treatment was supplied by the sponsor and accountability was reconciled at each monitoring visit.

9.5 Efficacy and safety variables

Efficacy was assessed according to RECIST version 1.1. Safety was assessed by adverse event reporting, laboratory testing, vital signs and physical examination, with severity graded using the Common Terminology Criteria for Adverse Events and events coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 22.0; concomitant medications were coded using the WHO Drug Dictionary (WHODrug) Global B2, March 2019. Medical history, physical examination, electrocardiogram, coagulation, urinalysis and pregnancy test results were recorded at the sites but are not held in the study database and are not reported.

9.6 Data quality assurance

Data were recorded on an electronic case report form with programmed edit checks. Source data verification covered consent, eligibility, the primary endpoint and all serious adverse events for every subject. Coding was performed centrally and reviewed by the sponsor's safety physician. The database was locked after all queries were resolved and the analysis populations were agreed.

9.7 Statistical methods planned in the protocol

No formal hypothesis testing is planned. Sample size follows the operating characteristics of the 3+3 design and the number of subjects required to characterise safety at the selected dose. Proportions are presented with exact Clopper-Pearson 95 per cent confidence intervals and time-to-event endpoints estimated by the Kaplan-Meier method.

9.8 Changes in the conduct of the study or the planned analyses

The protocol was first approved as version 1.0 on 17 October 2016. It was amended once before the data cut-off of 3 April 2019:

Amendment 1 (protocol version 2.0, 1 September 2017): Clarification of eligibility and monitoring. Effect on the analysis, as stated in the amendment: the primary analysis is unchanged. No change to the analysis populations or to the primary endpoint is introduced by this amendment.

The protocol of record for this report is version 2.0 (amendment 1, 1 September 2017).

Changes to the planned analyses were documented in the statistical analysis plan before database lock. No analysis reported here was devised after the data were examined.

10 STUDY SUBJECTS

10.1 Disposition of subjects

58 subjects were enrolled and treated. Disposition is in Table 10-1. Of these, 41 discontinued study treatment, most commonly for disease progression.

Table 10-1 Subject Disposition – Safety Analysis Set

Disposition	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Subjects treated, n	6	9	31	12	58
Treatment ongoing at data cut-off, n (%)	3 (50.0)	2 (22.2)	8 (25.8)	4 (33.3)	17 (29.3)
Discontinued study treatment, n (%)	3 (50.0)	7 (77.8)	23 (74.2)	8 (66.7)	41 (70.7)
Disease progression	1 (16.7)	3 (33.3)	7 (22.6)	3 (25.0)	14 (24.1)
Adverse event	0 (0.0)	2 (22.2)	1 (3.2)	1 (8.3)	4 (6.9)
Death	1 (16.7)	0 (0.0)	4 (12.9)	1 (8.3)	6 (10.3)
Withdrawal by subject	1 (16.7)	1 (11.1)	5 (16.1)	1 (8.3)	8 (13.8)
Physician decision	0 (0.0)	1 (11.1)	2 (6.5)	2 (16.7)	5 (8.6)
Lost to follow-up	0 (0.0)	0 (0.0)	3 (9.7)	0 (0.0)	3 (5.2)
Protocol deviation	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)

Percentages are based on the number of subjects treated within each group.

10.1.1 Reasons for discontinuation

Disease progression. Recorded for 14 subjects (24.1 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Withdrawal by subject. Recorded for 8 subjects (13.8 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Death. Recorded for 6 subjects (10.3 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Physician decision. Recorded for 5 subjects (8.6 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Adverse event. Recorded for 4 subjects (6.9 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

Lost to follow-up. Recorded for 3 subjects (5.2 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of

treatment visit and the safety follow-up visit unless consent was withdrawn.

Protocol deviation. Recorded for 1 subjects (1.7 per cent of those treated). The date and the reason were recorded at the time the decision was taken, and each subject attended the end of treatment visit and the safety follow-up visit unless consent was withdrawn.

10.2 Protocol deviations

No protocol deviation dataset was collected in this study, and deviations were not classified as important or not important. The deviations held in the data are those recorded as the reason for discontinuing study treatment: 1 subject (1.7 per cent of those treated, Table 10-1). This subject is listed at Appendix 16.2.2. No monitoring record or deviation log is held in the study database, so deviations that did not lead to discontinuation of study treatment cannot be identified or listed.

11 EFFICACY EVALUATION

11.1 Data sets analysed

Table 11-1 Analysis sets

Analysis set	Subjects	Definition
Enrolled	58	All subjects assigned to treatment.
Safety	58	All subjects who received at least one dose.
Full analysis	55	All subjects in the safety analysis set with at least one post-baseline disease assessment.
Efficacy evaluable	55	Subjects with at least one evaluable post-baseline assessment.

11.2 Demographic and other baseline characteristics

Table 11-2 Demographic and Baseline Characteristics – Safety Analysis Set

Characteristic	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Age (years)					
Mean (SD)	65.8 (2.04)	66.2 (10.96)	66.4 (8.86)	66.6 (9.79)	66.3 (8.78)
Median	66.0	65.0	68.0	67.5	67.0
Min, max	63, 68	50, 86	50, 84	47, 80	47, 86
>= 65 years, n (%)	4 (66.7)	5 (55.6)	20 (64.5)	8 (66.7)	37 (63.8)
Sex, n (%)					
Male	3 (50.0)	3 (33.3)	21 (67.7)	10 (83.3)	37 (63.8)
Female	3 (50.0)	6 (66.7)	10 (32.3)	2 (16.7)	21 (36.2)
Race, n (%)					
White	6 (100.0)	5 (55.6)	26 (83.9)	7 (58.3)	44 (75.9)
Asian	0 (0.0)	1 (11.1)	1 (3.2)	3 (25.0)	5 (8.6)
Black or African American	0 (0.0)	2 (22.2)	3 (9.7)	1 (8.3)	6 (10.3)
Other	0 (0.0)	1 (11.1)	1 (3.2)	1 (8.3)	3 (5.2)
ECOG performance status, n (%)					
0	3 (50.0)	4 (44.4)	15 (48.4)	6 (50.0)	28 (48.3)
1	2 (33.3)	4 (44.4)	12 (38.7)	5 (41.7)	23 (39.7)
2	1 (16.7)	1 (11.1)	4 (12.9)	1 (8.3)	7 (12.1)
BRCA mutation status, n (%)					
BRCA1 mutated	0 (0.0)	2 (22.2)	10 (32.3)	3 (25.0)	15 (25.9)
BRCA2 mutated	3 (50.0)	0 (0.0)	7 (22.6)	2 (16.7)	12 (20.7)
BRCA wild-type	3 (50.0)	7 (77.8)	14 (45.2)	7 (58.3)	31 (53.4)
Homologous recombination deficiency, n (%)	3 (50.0)	4 (44.4)	17 (54.8)	8 (66.7)	32 (55.2)
Number of prior lines of therapy					
Median	1.5	1.0	2.0	1.0	1.0
>= 3, n (%)	2 (33.3)	0 (0.0)	6 (19.4)	1 (8.3)	9 (15.5)

SD = standard deviation; ECOG = Eastern Cooperative Oncology Group. Percentages are based on the number of subjects in the safety analysis set within each group.

11.2.1 Discussion of each baseline characteristic

Age. The median age was 67 years, with a range of 47 to 86. 37 subjects (63.8 per cent) were aged 65 or over. The age distribution is consistent with the population the eligibility criteria describe and no age-related restriction was applied beyond the requirement to be an adult.

Sex. 37 subjects (63.8 per cent) were male and 21 (36.2 per cent) female. No eligibility criterion selected on sex, so the distribution reflects the population presenting with this disease at this point in the pathway.

Race. White: 44 (75.9 per cent); Black or African American: 6 (10.3 per cent); Asian: 5 (8.6 per cent); Other: 3 (5.2 per cent). Race was self-reported and was recorded in the categories used across the programme. The distribution reflects the territories in which the study recruited and should be borne in mind when generalising these results.

Performance status at baseline. score 0: 28 (48.3 per cent); score 1: 23 (39.7 per cent); score 2: 7 (12.1 per cent). Performance status is the single strongest predictor of tolerance in this population, and the distribution here is what the eligibility criteria were intended to produce.

Prior therapy. The median number of prior lines was 1, with a range of 1 to 4. Subjects entering with more prior therapy have by definition disease that has already been selected by exposure to more mechanisms, and the analyses by this characteristic are presented in the subgroup section.

BRCA mutation status. BRCA wild-type: 31 (53.4 per cent); BRCA1 mutated: 15 (25.9 per cent); BRCA2 mutated: 12 (20.7 per cent). Recorded at baseline and prespecified as a subgroup.

Homologous recombination status. Present in 32 subjects (55.2 per cent) at baseline. This factor was prespecified as a subgroup and is reported in the subgroup analyses.

11.3 Measurements of treatment compliance

The median duration of exposure was 122 days and the median relative dose intensity was 84.2 per cent. Exposure is summarised in Table 11-3.

Table 11-3 Extent of Exposure – Safety Analysis Set

Exposure parameter	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Duration of treatment (days)					
Median	171.0	117.0	118.0	147.0	121.5
Q1, Q3	107.8, 198.2	74.0, 129.0	83.5, 156.0	110.5, 226.2	86.0, 164.8
Min, max	65, 236	8, 308	8, 238	8, 448	8, 448
Cycles started, median	6.0	4.0	4.0	5.5	4.0
Relative dose intensity (%), median	83.8	83.9	84.7	82.9	84.2
Any dose reduction, n (%)	0 (0.0)	2 (22.2)	6 (19.4)	4 (33.3)	12 (20.7)
Any dose interruption, n (%)	1 (16.7)	3 (33.3)	5 (16.1)	1 (8.3)	10 (17.2)

Q1 = first quartile; Q3 = third quartile. Relative dose intensity is the total dose received divided by the total dose planned over the treatment period.

All 58 subjects who were enrolled received at least one dose and are included in the safety population. The median duration of exposure was 122 days, with a range of 8 to 448.

The median relative dose intensity was 84.2 per cent. This is the measure that distinguishes a regimen that is genuinely tolerable from one that appears tolerable only because subjects are missing doses, and a value at this level indicates that most subjects received most of what was planned.

12 subjects (20.7 per cent) had at least one dose reduction and 10 (17.2 per cent) had at least one interruption. Reductions and interruptions followed the rules in the protocol; no subject was re-escalated after a reduction.

Exposure was sufficient to characterise both the efficacy and the safety of the regimen. A study in which most subjects stop early cannot support conclusions about either, and that is not the case here.

11.4 Efficacy results

11.4.1 Analysis of the primary endpoint

The primary endpoint was the incidence of dose-limiting toxicity and the maximum tolerated dose derived from it; results are presented in Section 11.4.2.1.

Table 11-4 Best Overall Response by RECIST 1.1 – Full Analysis Set

Best overall response	100 mg (N=6)	200 mg (N=8)	300 mg (N=30)	400 mg (N=11)	Total (N=55)
Complete response (CR)	0 (0.0)	1 (12.5)	2 (6.7)	1 (9.1)	4 (7.3)
Partial response (PR)	0 (0.0)	2 (25.0)	6 (20.0)	4 (36.4)	12 (21.8)
Stable disease	4 (66.7)	1 (12.5)	13 (43.3)	3 (27.3)	21 (38.2)
Progressive disease	2 (33.3)	4 (50.0)	9 (30.0)	3 (27.3)	18 (32.7)
Not evaluable	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Objective response rate, n (%)	0 (0.0)	3 (37.5)	8 (26.7)	5 (45.5)	16 (29.1)
95% CI	(0.0, 45.9)	(8.5, 75.5)	(12.3, 45.9)	(16.7, 76.6)	(17.6, 42.9)

CI = confidence interval. Assessed according to RECIST version 1.1. Confidence intervals are exact (Clopper-Pearson).

Primary endpoint. The primary endpoint was the incidence of dose-limiting toxicity and the maximum tolerated dose derived from it; results are presented in Section 11.4.2.1. The estimate and its confidence interval are in Table 11-4. The interval rather than the point estimate is what

should be read, because it is the interval that states how precisely this study has answered the question.

Response. 16 of 55 subjects (29.1 per cent, 95 per cent confidence interval (17.6, 42.9)) had a response. Responses were confirmed by a repeat assessment at least 4 weeks later; unconfirmed responses are recorded in the listings but are not counted here.

Duration of response. The median was NE months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Progression-free survival. The median was 8.5 months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Overall survival. The median was NE months. Follow-up was still accruing at the data cut-off, so this estimate is immature and will change as further events are observed.

Consistency. The direction of effect was consistent across the prespecified subgroups reported above. None of those comparisons was powered and none is adjusted for multiplicity, so a difference between subgroups is not evidence of a difference in effect.

11.4.2 Analysis of time-to-event endpoints

Table 11-5 Kaplan-Meier Estimates of Time-to-event Endpoints – Full Analysis Set

Endpoint	100 mg (N=6)	200 mg (N=8)	300 mg (N=30)	400 mg (N=11)	Total (N=55)
Duration of response, median (months)	NE	NE	9.8	NE	NE
25th, 75th percentile	NE	NE	(9.8, NE)	(7.1, NE)	(9.8, NE)
Progression-free survival, median (months)	3.1	12.4	7.1	8.7	8.5
25th, 75th percentile	(2.9, 3.3)	(9.7, 12.7)	(3.0, 15.9)	(2.5, NE)	(3.0, 12.7)
Overall survival, median (months)	NE	NE	NE	NE	NE
25th, 75th percentile	NE	(12.7, NE)	NE	NE	NE

NE = not estimable. Duration of response is analysed among responding subjects only. It is measured from the first dose to the end of response recorded in the analysis dataset; response and progression dates cannot be re-derived because no assessment-level tumour data were collected in the database.

11.4.2.1 Dose-limiting toxicity and dose determination

Dose-limiting toxicity was reported in 11 of 53 evaluable subjects. Escalation decisions are summarised below. The maximum tolerated dose was 300 mg orally twice daily on a continuous schedule; the next higher dose exceeded it.

Table 11-DLT Dose-limiting Toxicity and Escalation Decisions by Cohort – DLT-evaluable Set

Dose cohort	Subjects treated	DLT-evaluable	Subjects with DLT	DLT terms	Escalation decision
100 mg	6	6	0 (0.0)	None	Escalate
200 mg	9	7	2 (28.6)	Anaemia (1); Headache (1)	Escalate
300 mg	31	28	5 (17.9)	Anaemia (2); Neutropenia (2); Leukopenia (1); Lymphopenia (1); Thrombocytopenia (1)	Expand; declared RP2D
400 mg	12	12	4 (33.3)	Neutropenia (3); Anaemia (1); Lymphopenia (1)	MTD exceeded; de-escalate

DLT = dose-limiting toxicity; MTD = maximum tolerated dose; RP2D = recommended phase 2 dose. The DLT-evaluable set comprises subjects who received at least 21 days of study treatment or experienced a DLT.

11.4.3 Subgroup analyses

The primary endpoint was examined in the prespecified subgroups below. These analyses were not powered and are presented to describe consistency of effect; no conclusion is drawn from any single subgroup.

Table 11-6 Overall Response Rate and Progression-free Survival by BRCA Mutation Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
BRCA wild-type	28	9 (32.1)	(15.9, 52.4)	8.7
BRCA1 mutated	15	6 (40.0)	(16.3, 67.7)	5.8
BRCA2 mutated	12	1 (8.3)	(0.2, 38.5)	5.4

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-7 Overall Response Rate and Progression-free Survival by Homologous Recombination Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Negative	23	4 (17.4)	(5.0, 38.8)	8.5
Positive	32	12 (37.5)	(21.1, 56.3)	6.7

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-8 Overall Response Rate and Progression-free Survival by Prior Lines of Therapy – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
1 to 2	46	13 (28.3)	(16.0, 43.5)	8.7
>= 3	9	3 (33.3)	(7.5, 70.1)	4.8

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-9 Overall Response Rate and Progression-free Survival by Age Category – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
< 65 years	20	5 (25.0)	(8.7, 49.1)	8.9
>= 65 years	35	11 (31.4)	(16.9, 49.3)	7.1

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-10 Overall Response Rate and Progression-free Survival by Sex – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Female	19	5 (26.3)	(9.1, 51.2)	9.7
Male	36	11 (30.6)	(16.3, 48.1)	5.9

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 11-11 Overall Response Rate and Progression-free Survival by ECOG Performance Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
0	28	6 (21.4)	(8.3, 41.0)	5.9
1	21	10 (47.6)	(25.7, 70.2)	8.9
2	6	0 (0.0)	(0.0, 45.9)	3.5

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

BRCA Mutation Status. BRCA wild-type: 9 of 28 (32.1 per cent); BRCA1 mutated: 6 of 15 (40.0 per cent); BRCA2 mutated: 1 of 12 (8.3 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Homologous Recombination Status. Negative: 4 of 23 (17.4 per cent); Positive: 12 of 32 (37.5 per cent). The subgroups are of unequal size and none of these comparisons was powered, so

the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Prior Lines of Therapy. 1 to 2: 13 of 46 (28.3 per cent); >= 3: 3 of 9 (33.3 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Age Category. < 65 years: 5 of 20 (25.0 per cent); >= 65 years: 11 of 35 (31.4 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

Sex. Female: 5 of 19 (26.3 per cent); Male: 11 of 36 (30.6 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

ECOG Performance Status. 0: 6 of 28 (21.4 per cent); 1: 10 of 21 (47.6 per cent); 2: 0 of 6 (0.0 per cent). The subgroups are of unequal size and none of these comparisons was powered, so the differences between them are presented to show consistency of direction rather than to support a conclusion about any one subgroup.

11.4.4 Efficacy conclusions

The maximum tolerated dose was 300 mg orally twice daily: dose-limiting toxicities occurred in 0 of 6, 2 of 7, 5 of 28 and 4 of 12 DLT-evaluable subjects at 100, 200, 300 and 400 mg, and 300 mg is the recommended phase 2 dose. Kelvaparib produced antitumor activity in advanced solid tumours with a homologous recombination repair defect (overall response rate 29.1 per cent); efficacy observations are exploratory.

11.5 By-subject efficacy data

Individual response data for every subject are listed at Appendix 16.2.7, and individual time-to-event data at Appendix 16.2.8.

12 SAFETY EVALUATION

12.1 Extent of exposure

All 58 subjects received at least one dose. The extent of exposure is in Table 11-3. Total exposure was 21.2 patient-years, a mean of 0.366 patient-years per subject, and it was not evenly distributed across the dose cohorts: from 0.313 patient-years per subject at 300 mg to 0.486 at 400 mg. Exposure expressed as person-time, and adverse event counts expressed per unit of person-time, are given at Table 14.3.16.

12.2 Adverse events

12.2.1 Overview

Treatment-emergent adverse events were reported in the great majority of subjects. 41 subjects (70.7 per cent) had an event of grade 3 or higher, 22 (37.9 per cent) had a serious event and 10 (17.2 per cent) discontinued study treatment because of an event.

Table 12-1 Overview of Treatment-emergent Adverse Events – Safety Analysis Set

Category, n (%)	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Any TEAE	6 (100.0)	9 (100.0)	31 (100.0)	12 (100.0)	58 (100.0)
Any treatment-related TEAE	5 (83.3)	9 (100.0)	31 (100.0)	11 (91.7)	56 (96.6)
Any grade >= 3 TEAE	3 (50.0)	7 (77.8)	21 (67.7)	10 (83.3)	41 (70.7)
Any serious TEAE	1 (16.7)	2 (22.2)	12 (38.7)	7 (58.3)	22 (37.9)
Any TEAE leading to dose reduction	0 (0.0)	2 (22.2)	6 (19.4)	4 (33.3)	12 (20.7)
Any TEAE leading to discontinuation	1 (16.7)	2 (22.2)	5 (16.1)	2 (16.7)	10 (17.2)
Any fatal TEAE	1 (16.7)	0 (0.0)	4 (12.9)	1 (8.3)	6 (10.3)

TEAE = treatment-emergent adverse event. A subject is counted once within each category.

12.2.1.1 Discussion of each event term

Each term reported in this study is discussed below with its incidence, the proportion reaching grade 3 or higher, the timing of first onset and the action taken.

Nausea was reported in 37 subjects (63.8 per cent), of whom 1 (1.7 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 24 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 8 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Anaemia was reported in 35 subjects (60.3 per cent), of whom 18 (31.0 per cent of the study population) had an event of grade 3 or higher, and 9 had an event recorded as serious. The median time to first onset was 40 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 7 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Fatigue was reported in 33 subjects (56.9 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 23 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Decreased appetite was reported in 24 subjects (41.4 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 45 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 4 subjects had an event that had not resolved at the data cut-off. The event falls within Metabolism and nutrition disorders.

Constipation was reported in 22 subjects (37.9 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 42 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Vomiting was reported in 18 subjects (31.0 per cent), of whom 1 (1.7 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 34 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 4 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Abdominal pain was reported in 17 subjects (29.3 per cent), of whom 1 (1.7 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 46 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Neutropenia was reported in 17 subjects (29.3 per cent), of whom 8 (13.8 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 17 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 5 subjects had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Thrombocytopenia was reported in 16 subjects (27.6 per cent), of whom 6 (10.3 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 22 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. 1 subject had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Headache was reported in 15 subjects (25.9 per cent), of whom 1 (1.7 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 44 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Diarrhoea was reported in 13 subjects (22.4 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 44 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Dyspepsia was reported in 12 subjects (20.7 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 44 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Gastrointestinal disorders.

Urinary tract infection was reported in 11 subjects (19.0 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 51 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within Infections and infestations.

Asthenia was reported in 11 subjects (19.0 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 30 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Insomnia was reported in 11 subjects (19.0 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 1 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Psychiatric disorders.

Lymphopenia was reported in 10 subjects (17.2 per cent), of whom 3 (5.2 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 42 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. It led to permanent discontinuation of study treatment in 1 subject. The event falls within Blood and lymphatic system disorders.

Back pain was reported in 9 subjects (15.5 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 31 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

Nasopharyngitis was reported in 9 subjects (15.5 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 8 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within Infections and infestations.

Dizziness was reported in 9 subjects (15.5 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 30 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within Nervous system disorders.

Pyrexia was reported in 9 subjects (15.5 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 22 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within General disorders and administration site conditions.

Alanine aminotransferase increased was reported in 9 subjects (15.5 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 43 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Investigations.

Aspartate aminotransferase increased was reported in 8 subjects (13.8 per cent), of whom 1 (1.7 per cent of the study population) had an event of grade 3 or higher, and 1 had an event recorded as serious. The median time to first onset was 22 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Investigations.

Arthralgia was reported in 8 subjects (13.8 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 43 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Musculoskeletal and connective tissue disorders.

Hypertension was reported in 8 subjects (13.8 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 37 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. The event falls within Vascular disorders.

Dysgeusia was reported in 7 subjects (12.1 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 12 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. The event falls within Nervous system disorders.

Weight decreased was reported in 7 subjects (12.1 per cent), of whom 0 (0.0 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 19 days from the first dose, so the event arose within the period of most intensive monitoring in the

protocol. 2 subjects had an event that had not resolved at the data cut-off. The event falls within Investigations.

Leukopenia was reported in 6 subjects (10.3 per cent), of whom 3 (5.2 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 41 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 1 subject had an event that had not resolved at the data cut-off. The event falls within Blood and lymphatic system disorders.

Dyspnoea was reported in 6 subjects (10.3 per cent), of whom 1 (1.7 per cent of the study population) had an event of grade 3 or higher. The median time to first onset was 33 days from the first dose, so the event arose within the period of most intensive monitoring in the protocol. 3 subjects had an event that had not resolved at the data cut-off. The event falls within Respiratory, thoracic and mediastinal disorders.

12.2.2 Events by system organ class and preferred term

Table 12-2 Treatment-emergent Adverse Events Occurring in at Least 10% of Subjects by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Gastrointestinal disorders	5 (83.3)	9 (100.0)	30 (96.8)	10 (83.3)	54 (93.1)
Nausea	2 (33.3)	5 (55.6)	21 (67.7)	9 (75.0)	37 (63.8)
Constipation	1 (16.7)	2 (22.2)	16 (51.6)	3 (25.0)	22 (37.9)
Vomiting	1 (16.7)	3 (33.3)	12 (38.7)	2 (16.7)	18 (31.0)
Abdominal pain	3 (50.0)	4 (44.4)	7 (22.6)	3 (25.0)	17 (29.3)
Diarrhoea	3 (50.0)	2 (22.2)	7 (22.6)	1 (8.3)	13 (22.4)
Dyspepsia	1 (16.7)	2 (22.2)	6 (19.4)	3 (25.0)	12 (20.7)
Blood and lymphatic system disorders	4 (66.7)	8 (88.9)	23 (74.2)	10 (83.3)	45 (77.6)
Anaemia	2 (33.3)	4 (44.4)	19 (61.3)	10 (83.3)	35 (60.3)
Neutropenia	2 (33.3)	4 (44.4)	8 (25.8)	3 (25.0)	17 (29.3)
Thrombocytopenia	1 (16.7)	2 (22.2)	10 (32.3)	3 (25.0)	16 (27.6)
Lymphopenia	1 (16.7)	2 (22.2)	4 (12.9)	3 (25.0)	10 (17.2)
Leukopenia	1 (16.7)	1 (11.1)	2 (6.5)	2 (16.7)	6 (10.3)
General disorders and administration site conditions	4 (66.7)	6 (66.7)	25 (80.6)	8 (66.7)	43 (74.1)
Fatigue	4 (66.7)	4 (44.4)	18 (58.1)	7 (58.3)	33 (56.9)
Asthenia	0 (0.0)	2 (22.2)	6 (19.4)	3 (25.0)	11 (19.0)
Pyrexia	1 (16.7)	0 (0.0)	7 (22.6)	1 (8.3)	9 (15.5)
Nervous system disorders	2 (33.3)	3 (33.3)	16 (51.6)	5 (41.7)	26 (44.8)
Headache	0 (0.0)	1 (11.1)	10 (32.3)	4 (33.3)	15 (25.9)
Dizziness	2 (33.3)	1 (11.1)	6 (19.4)	0 (0.0)	9 (15.5)
Dysgeusia	1 (16.7)	1 (11.1)	4 (12.9)	1 (8.3)	7 (12.1)
Infections and infestations	2 (33.3)	4 (44.4)	14 (45.2)	5 (41.7)	25 (43.1)
Urinary tract infection	1 (16.7)	2 (22.2)	4 (12.9)	4 (33.3)	11 (19.0)
Nasopharyngitis	1 (16.7)	2 (22.2)	5 (16.1)	1 (8.3)	9 (15.5)
Metabolism and nutrition disorders	2 (33.3)	5 (55.6)	12 (38.7)	5 (41.7)	24 (41.4)
Decreased appetite	2 (33.3)	5 (55.6)	12 (38.7)	5 (41.7)	24 (41.4)
Investigations	1 (16.7)	4 (44.4)	13 (41.9)	6 (50.0)	24 (41.4)
Alanine aminotransferase increased	0 (0.0)	1 (11.1)	6 (19.4)	2 (16.7)	9 (15.5)
Aspartate aminotransferase increased	1 (16.7)	0 (0.0)	4 (12.9)	3 (25.0)	8 (13.8)
Weight decreased	0 (0.0)	2 (22.2)	3 (9.7)	2 (16.7)	7 (12.1)
Musculoskeletal and connective tissue disorders	0 (0.0)	4 (44.4)	13 (41.9)	3 (25.0)	20 (34.5)
Back pain	0 (0.0)	2 (22.2)	7 (22.6)	0 (0.0)	9 (15.5)
Arthralgia	0 (0.0)	1 (11.1)	5 (16.1)	2 (16.7)	8 (13.8)

System organ class / Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Respiratory, thoracic and mediastinal disorders	0 (0.0)	1 (11.1)	8 (25.8)	3 (25.0)	12 (20.7)
Dyspnoea	0 (0.0)	1 (11.1)	3 (9.7)	2 (16.7)	6 (10.3)
Cough	0 (0.0)	0 (0.0)	5 (16.1)	1 (8.3)	6 (10.3)
Psychiatric disorders	2 (33.3)	1 (11.1)	7 (22.6)	1 (8.3)	11 (19.0)
Insomnia	2 (33.3)	1 (11.1)	7 (22.6)	1 (8.3)	11 (19.0)
Vascular disorders	0 (0.0)	1 (11.1)	7 (22.6)	0 (0.0)	8 (13.8)
Hypertension	0 (0.0)	1 (11.1)	7 (22.6)	0 (0.0)	8 (13.8)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 22.0.

Gastrointestinal disorders. Events in this class were reported in 54 subjects (93.1 per cent), and in 5 (8.6 per cent) at grade 3 or higher. The terms reported most often were nausea, constipation, vomiting. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Blood and lymphatic system disorders. Events in this class were reported in 45 subjects (77.6 per cent), and in 31 (53.4 per cent) at grade 3 or higher. The terms reported most often were anaemia, neutropenia, thrombocytopenia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

General disorders and administration site conditions. Events in this class were reported in 43 subjects (74.1 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were fatigue, asthenia, pyrexia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Nervous system disorders. Events in this class were reported in 26 subjects (44.8 per cent), and in 1 (1.7 per cent) at grade 3 or higher. The terms reported most often were headache, dizziness, dysgeusia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Infections and infestations. Events in this class were reported in 25 subjects (43.1 per cent), and in 6 (10.3 per cent) at grade 3 or higher. The terms reported most often were urinary tract infection, nasopharyngitis, pneumonia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Metabolism and nutrition disorders. Events in this class were reported in 24 subjects (41.4 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were decreased appetite. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Investigations. Events in this class were reported in 24 subjects (41.4 per cent), and in 4 (6.9 per cent) at grade 3 or higher. The terms reported most often were alanine aminotransferase increased, aspartate aminotransferase increased, weight decreased. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Musculoskeletal and connective tissue disorders. Events in this class were reported in 20 subjects (34.5 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were back pain, arthralgia, myalgia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Respiratory, thoracic and mediastinal disorders. Events in this class were reported in 12 subjects (20.7 per cent), and in 2 (3.4 per cent) at grade 3 or higher. The terms reported most often were dyspnoea, cough. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Psychiatric disorders. Events in this class were reported in 11 subjects (19.0 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were insomnia. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Vascular disorders. Events in this class were reported in 8 subjects (13.8 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were hypertension. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Skin and subcutaneous tissue disorders. Events in this class were reported in 4 subjects (6.9 per cent), and in 0 (0.0 per cent) at grade 3 or higher. The terms reported most often were rash. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

Neoplasms benign, malignant and unspecified. Events in this class were reported in 4 subjects (6.9 per cent), and in 4 (6.9 per cent) at grade 3 or higher. The terms reported most often were myelodysplastic syndrome, malignant neoplasm progression. The pattern is consistent with what is anticipated for oral PARP1-selective inhibitor in this population.

12.2.3 Events of grade 3 or higher

Table 12-3 Treatment-emergent Adverse Events of Grade 3 or Higher by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Blood and lymphatic system disorders	2 (33.3)	4 (44.4)	16 (51.6)	9 (75.0)	31 (53.4)
Anaemia	1 (16.7)	1 (11.1)	10 (32.3)	6 (50.0)	18 (31.0)
Neutropenia	0 (0.0)	2 (22.2)	3 (9.7)	3 (25.0)	8 (13.8)
Thrombocytopenia	0 (0.0)	1 (11.1)	4 (12.9)	1 (8.3)	6 (10.3)
Infections and infestations	0 (0.0)	1 (11.1)	5 (16.1)	0 (0.0)	6 (10.3)
Gastrointestinal disorders	2 (33.3)	1 (11.1)	2 (6.5)	0 (0.0)	5 (8.6)
Neoplasms benign, malignant and unspecified	0 (0.0)	1 (11.1)	2 (6.5)	1 (8.3)	4 (6.9)
Investigations	0 (0.0)	2 (22.2)	2 (6.5)	0 (0.0)	4 (6.9)
Respiratory, thoracic and mediastinal disorders	0 (0.0)	0 (0.0)	2 (6.5)	0 (0.0)	2 (3.4)
Nervous system disorders	0 (0.0)	1 (11.1)	0 (0.0)	0 (0.0)	1 (1.7)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 22.0.

12.2.4 Events considered related to study treatment

Table 12-4 Treatment-related Treatment-emergent Adverse Events Occurring in at Least 5% of Subjects – Safety Analysis Set

Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Fatigue	3 (50.0)	2 (22.2)	12 (38.7)	4 (33.3)	21 (36.2)
Anaemia	0 (0.0)	2 (22.2)	13 (41.9)	6 (50.0)	21 (36.2)
Decreased appetite	1 (16.7)	5 (55.6)	8 (25.8)	2 (16.7)	16 (27.6)
Nausea	1 (16.7)	1 (11.1)	9 (29.0)	5 (41.7)	16 (27.6)
Constipation	0 (0.0)	1 (11.1)	10 (32.3)	2 (16.7)	13 (22.4)
Abdominal pain	3 (50.0)	3 (33.3)	5 (16.1)	1 (8.3)	12 (20.7)
Vomiting	0 (0.0)	3 (33.3)	7 (22.6)	1 (8.3)	11 (19.0)
Insomnia	2 (33.3)	1 (11.1)	6 (19.4)	1 (8.3)	10 (17.2)
Urinary tract infection	1 (16.7)	2 (22.2)	4 (12.9)	3 (25.0)	10 (17.2)
Neutropenia	0 (0.0)	1 (11.1)	5 (16.1)	3 (25.0)	9 (15.5)
Headache	0 (0.0)	1 (11.1)	6 (19.4)	1 (8.3)	8 (13.8)
Thrombocytopenia	0 (0.0)	2 (22.2)	4 (12.9)	2 (16.7)	8 (13.8)
Lymphopenia	1 (16.7)	1 (11.1)	3 (9.7)	1 (8.3)	6 (10.3)
Diarrhoea	1 (16.7)	1 (11.1)	3 (9.7)	1 (8.3)	6 (10.3)
Asthenia	0 (0.0)	1 (11.1)	3 (9.7)	2 (16.7)	6 (10.3)
Weight decreased	0 (0.0)	2 (22.2)	2 (6.5)	2 (16.7)	6 (10.3)
Dyspepsia	1 (16.7)	1 (11.1)	3 (9.7)	0 (0.0)	5 (8.6)
Leukopenia	1 (16.7)	1 (11.1)	1 (3.2)	2 (16.7)	5 (8.6)
Nasopharyngitis	0 (0.0)	2 (22.2)	2 (6.5)	1 (8.3)	5 (8.6)
Alanine aminotransferase increased	0 (0.0)	1 (11.1)	2 (6.5)	2 (16.7)	5 (8.6)
Arthralgia	0 (0.0)	1 (11.1)	2 (6.5)	2 (16.7)	5 (8.6)
Pyrexia	0 (0.0)	0 (0.0)	4 (12.9)	1 (8.3)	5 (8.6)
Dysgeusia	1 (16.7)	1 (11.1)	2 (6.5)	0 (0.0)	4 (6.9)
Dizziness	1 (16.7)	0 (0.0)	3 (9.7)	0 (0.0)	4 (6.9)
Back pain	0 (0.0)	1 (11.1)	3 (9.7)	0 (0.0)	4 (6.9)
Hypertension	0 (0.0)	1 (11.1)	3 (9.7)	0 (0.0)	4 (6.9)
Rash	1 (16.7)	1 (11.1)	1 (3.2)	0 (0.0)	3 (5.2)
Myalgia	0 (0.0)	1 (11.1)	1 (3.2)	1 (8.3)	3 (5.2)
Dyspnoea	0 (0.0)	1 (11.1)	1 (3.2)	1 (8.3)	3 (5.2)

Relationship assessed by the investigator. Percentages are based on the number of subjects treated in each group. Overall N = 58.

12.2.5 Timing of onset

Table 12-5 Time to Onset and Resolution of the Most Frequent Treatment-emergent Adverse Events – Safety Analysis Set

Preferred term	Events, n	Median time to onset (days)	Resolved, n (%)	Median time to resolution (days)
Nausea	37	24.0	29 (78.4)	9.0
Anaemia	35	40.0	28 (80.0)	14.0
Fatigue	33	23.0	31 (93.9)	9.0
Decreased appetite	24	44.5	20 (83.3)	12.0
Constipation	22	39.0	19 (86.4)	8.0
Vomiting	18	23.0	14 (77.8)	5.5
Abdominal pain	17	46.0	15 (88.2)	10.0
Neutropenia	17	17.0	12 (70.6)	9.0
Thrombocytopenia	16	19.0	15 (93.8)	7.0
Headache	15	44.0	12 (80.0)	16.5
Diarrhoea	13	44.0	12 (92.3)	8.0
Dyspepsia	12	22.5	10 (83.3)	10.5
Insomnia	11	1.0	10 (90.9)	9.5
Urinary tract infection	11	51.0	8 (72.7)	15.5
Asthenia	11	30.0	8 (72.7)	8.5
Lymphopenia	10	38.0	10 (100.0)	17.0
Nasopharyngitis	9	8.0	6 (66.7)	12.0
Dizziness	9	30.0	6 (66.7)	9.5
Pyrexia	9	22.0	7 (77.8)	13.0
Alanine aminotransferase increased	9	43.0	8 (88.9)	6.5

Events are counted at the event level rather than the subject level. NE = not estimable.

12.2.6 Adverse events of special interest

Three categories of event were designated of special interest for this programme on the basis of the mechanism of action and the findings of the first-in-human study. Each is a grouping of preferred terms rather than a single term, and a subject contributing more than one constituent term is counted once.

Table 12-6 Adverse Events of Special Interest - Safety Analysis Set

Event of special interest	Minimum grade	Subjects, n (%)	Grade >= 3, n (%)	Serious, n (%)
Myelodysplastic syndrome or acute myeloid leukaemia	Any	2 (3.4)	2 (3.4)	1 (1.7)
Pneumonitis	Any	0 (0.0)	0 (0.0)	0 (0.0)
Severe anaemia	3	18 (31.0)	18 (31.0)	9 (15.5)

Constituent preferred terms. Myelodysplastic syndrome or acute myeloid leukaemia: Myelodysplastic syndrome; Pneumonitis: Pneumonitis; Severe anaemia: Anaemia. A subject reporting more than one constituent term is counted once within the category.

Myelodysplastic syndrome or acute myeloid leukaemia

Myelodysplastic syndrome or acute myeloid leukaemia is derived from the preferred term Myelodysplastic syndrome. The category was reported in 2 of 58 subjects (3.4 per cent), of whom 2 (3.4 per cent) had an event of grade 3 or higher and 0 had a fatal outcome.

The median time to first onset was 111 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Pneumonitis

Pneumonitis is reported as the coded preferred term. The category was reported in 0 of 58 subjects (0.0 per cent), of whom 0 (0.0 per cent) had an event of grade 3 or higher and 0 had a fatal outcome.

The median time to first onset was not estimable, no event having been reported. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Severe anaemia

Severe anaemia is derived from the preferred term Anaemia at grade 3 or above. The category was reported in 18 of 58 subjects (31.0 per cent), of whom 18 (31.0 per cent) had an event of grade 3 or higher and 0 had a fatal outcome.

The median time to first onset was 45 days from the first dose. Events were managed by the measures set out in the protocol; no subject was withdrawn from the study for this reason without the event first being reported to the sponsor within 24 hours.

Nausea was the most frequent event overall, reported in 37 subjects (63.8 per cent), of whom 1 (1.7 per cent) had an event of grade 3 or higher.

12.3 Deaths, other serious adverse events and other significant events

12.3.1 Deaths

8 subjects died. The causes recorded are in Table 12-7. Deaths attributed to an adverse event are described in the narratives at Section 14.3.3.

Table 12-7 Deaths – Safety Analysis Set

Category, n (%)	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Any death	1 (16.7)	1 (11.1)	5 (16.1)	1 (8.3)	8 (13.8)
Death within 30 days of last dose	1 (16.7)	0 (0.0)	4 (12.9)	1 (8.3)	6 (10.3)
Primary cause of death					
Disease progression	0 (0.0)	1 (11.1)	0 (0.0)	0 (0.0)	1 (1.7)
Adverse event	1 (16.7)	0 (0.0)	4 (12.9)	1 (8.3)	6 (10.3)
Other	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)

Percentages are based on the number of subjects treated within each group.

8 of 58 subjects (13.8 per cent) died. The primary cause recorded for each is below. Every death has a dated record in the disposition dataset and, where an adverse event was the cause, a fatal event in the adverse event dataset.

Disease progression. Recorded as the primary cause for 1 subjects (12.5 per cent of those who died). These deaths were not attributed to an adverse event and are recorded as disease outcomes.

Adverse event. Recorded as the primary cause for 6 subjects (75.0 per cent of those who died). Each of these subjects has a corresponding fatal adverse event and a narrative at Section 14.3.3.

Other. Recorded as the primary cause for 1 subjects (12.5 per cent of those who died). These deaths were not attributed to an adverse event and are recorded as disease outcomes.

12.3.2 Serious adverse events

Table 12-8 Serious Treatment-emergent Adverse Events by Preferred Term – Safety Analysis Set

Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Anaemia	0 (0.0)	1 (11.1)	4 (12.9)	4 (33.3)	9 (15.5)
Blood creatinine increased	0 (0.0)	2 (22.2)	0 (0.0)	0 (0.0)	2 (3.4)
Malignant neoplasm progression	0 (0.0)	0 (0.0)	1 (3.2)	1 (8.3)	2 (3.4)
Pneumonia	0 (0.0)	0 (0.0)	2 (6.5)	0 (0.0)	2 (3.4)
Ascites	1 (16.7)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.7)
Aspartate aminotransferase increased	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Cough	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Intestinal obstruction	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Lymphopenia	0 (0.0)	0 (0.0)	0 (0.0)	1 (8.3)	1 (1.7)
Myelodysplastic syndrome	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Nausea	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Neutropenia	0 (0.0)	0 (0.0)	0 (0.0)	1 (8.3)	1 (1.7)
Sepsis	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Thrombocytopenia	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)

Subjects are counted once per preferred term. Percentages are based on N = 58.

12.3.3 Events leading to discontinuation

Table 12-9 Treatment-emergent Adverse Events Leading to Discontinuation of Study Treatment – Safety Analysis Set

Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Malignant neoplasm progression	0 (0.0)	0 (0.0)	1 (3.2)	1 (8.3)	2 (3.4)
Sepsis	0 (0.0)	0 (0.0)	2 (6.5)	0 (0.0)	2 (3.4)
Ascites	1 (16.7)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.7)
Intestinal obstruction	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Lymphopenia	0 (0.0)	0 (0.0)	0 (0.0)	1 (8.3)	1 (1.7)
Myelodysplastic syndrome	0 (0.0)	1 (11.1)	0 (0.0)	0 (0.0)	1 (1.7)
Pneumonia	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Thrombocytopenia	0 (0.0)	1 (11.1)	0 (0.0)	0 (0.0)	1 (1.7)

Percentages are based on N = 58.

12.3.4 Narratives

Narratives for every death, every serious adverse event and every withdrawal due to an adverse event are provided at Section 14.3.3, bound separately.

12.4 Clinical laboratory evaluation

Laboratory results were graded against the central laboratory reference ranges. Shifts from baseline to the worst post-baseline grade are in Table 12-10. Individual laboratory data are listed at Appendix 16.2.10.

Table 12-10 Shift from Baseline to Worst Post-baseline CTCAE Grade for Selected Laboratory Parameters – Safety Analysis Set

Laboratory parameter	100 mg: worst grade 3 n (%)	100 mg: worst grade 4 n (%)	100 mg: shift >= 2 grades n (%)	200 mg: worst grade 3 n (%)	200 mg: worst grade 4 n (%)	200 mg: shift >= 2 grades n (%)	300 mg: worst grade 3 n (%)	300 mg: worst grade 4 n (%)	300 mg: shift >= 2 grades n (%)	400 mg: worst grade 3 n (%)	400 mg: worst grade 4 n (%)	400 mg: shift >= 2 grades n (%)	Total: worst grade 3 n (%)	Total: worst grade 4 n (%)	Total: shift >= 2 grades n (%)
Haemoglobin	0 (0.0)	0 (0.0)	2 (33.3)	1 (11.1)	2 (22.2)	4 (44.4)	10 (32.3)	4 (12.9)	17 (54.8)	3 (25.0)	0 (0.0)	5 (41.7)	14 (24.1)	6 (10.3)	28 (48.3)
Platelets	2 (33.3)	0 (0.0)	2 (33.3)	1 (11.1)	0 (0.0)	4 (44.4)	2 (6.5)	1 (3.2)	6 (19.4)	0 (0.0)	0 (0.0)	2 (16.7)	5 (8.6)	1 (1.7)	14 (24.1)
Neutrophils	0 (0.0)	0 (0.0)	2 (33.3)	1 (11.1)	1 (11.1)	3 (33.3)	4 (12.9)	0 (0.0)	4 (12.9)	1 (8.3)	0 (0.0)	1 (8.3)	6 (10.3)	1 (1.7)	10 (17.2)
Lymphocytes	0 (0.0)	0 (0.0)	0 (0.0)	1 (11.1)	0 (0.0)	0 (0.0)	4 (12.9)	1 (3.2)	6 (19.4)	2 (16.7)	0 (0.0)	2 (16.7)	7 (12.1)	1 (1.7)	8 (13.8)
CA-125	2 (33.3)	0 (0.0)	1 (16.7)	0 (0.0)	0 (0.0)	2 (22.2)	4 (12.9)	2 (6.5)	10 (32.3)	2 (16.7)	0 (0.0)	4 (33.3)	8 (13.8)	2 (3.4)	17 (29.3)
ALT	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	5 (16.1)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.7)	0 (0.0)	5 (8.6)
AST	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (11.1)	1 (11.1)	2 (6.5)	0 (0.0)	3 (9.7)	2 (16.7)	1 (8.3)	3 (25.0)	4 (6.9)	2 (3.4)	7 (12.1)
Creatinine	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	3 (9.7)	0 (0.0)	3 (9.7)	1 (8.3)	0 (0.0)	2 (16.7)	4 (6.9)	0 (0.0)	5 (8.6)
Lactate dehydrogenase	1 (16.7)	0 (0.0)	1 (16.7)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (3.2)	6 (19.4)	1 (8.3)	1 (8.3)	2 (16.7)	2 (3.4)	2 (3.4)	9 (15.5)
Albumin	0 (0.0)	0 (0.0)	1 (16.7)	0 (0.0)	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	2 (6.5)	1 (8.3)	1 (8.3)	2 (16.7)	2 (3.4)	1 (1.7)	5 (8.6)

CTCAE = Common Terminology Criteria for Adverse Events, version 5.0. Percentages are based on the number of subjects treated in each group. Overall N = 58.

12.4.1 Findings for each laboratory test

ALT. An abnormality of any grade was present at baseline in 16 subjects (27.6 per cent) and post-baseline in 24 (41.4 per cent). 9 subjects (15.5 per cent) shifted to a worse grade than their own baseline, and 1 (1.7 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

AST. An abnormality of any grade was present at baseline in 16 subjects (27.6 per cent) and post-baseline in 24 (41.4 per cent). 11 subjects (19.0 per cent) shifted to a worse grade than their own baseline, and 6 (10.3 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Albumin. An abnormality of any grade was present at baseline in 13 subjects (22.4 per cent) and post-baseline in 20 (34.5 per cent). 10 subjects (17.2 per cent) shifted to a worse grade than their own baseline, and 3 (5.2 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

CA-125. An abnormality of any grade was present at baseline in 19 subjects (32.8 per cent) and post-baseline in 49 (84.5 per cent). 43 subjects (74.1 per cent) shifted to a worse grade than their own baseline, and 10 (17.2 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Creatinine. An abnormality of any grade was present at baseline in 20 subjects (34.5 per cent) and post-baseline in 27 (46.6 per cent). 14 subjects (24.1 per cent) shifted to a worse grade than their own baseline, and 4 (6.9 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Haemoglobin. An abnormality of any grade was present at baseline in 23 subjects (39.7 per cent) and post-baseline in 50 (86.2 per cent). 44 subjects (75.9 per cent) shifted to a worse grade than their own baseline, and 20 (34.5 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Lactate dehydrogenase. An abnormality of any grade was present at baseline in 20 subjects (34.5 per cent) and post-baseline in 28 (48.3 per cent). 13 subjects (22.4 per cent) shifted to a worse grade than their own baseline, and 4 (6.9 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Lymphocytes. An abnormality of any grade was present at baseline in 19 subjects (32.8 per cent) and post-baseline in 32 (55.2 per cent). 23 subjects (39.7 per cent) shifted to a worse grade than their own baseline, and 8 (13.8 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Neutrophils. An abnormality of any grade was present at baseline in 21 subjects (36.2 per cent) and post-baseline in 36 (62.1 per cent). 26 subjects (44.8 per cent) shifted to a worse grade than their own baseline, and 7 (12.1 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

Platelets. An abnormality of any grade was present at baseline in 17 subjects (29.3 per cent) and post-baseline in 33 (56.9 per cent). 26 subjects (44.8 per cent) shifted to a worse grade than their own baseline, and 6 (10.3 per cent) reached grade 3 or higher at some point. Values were reviewed against the dose modification rules at every visit at which they were measured.

12.5 Vital signs, physical findings and other safety observations

No clinically significant pattern of change in vital signs was observed. Individual vital sign data are listed at Appendix 16.2.11.

12.5.1 Electrocardiograms

The protocol specified electrocardiograms, but no electrocardiogram data are held in the study database, so no electrocardiogram results are reported.

12.6 Safety conclusions

The safety profile observed was consistent with that anticipated for oral PARP1-selective inhibitor in this population. The events that occurred most often, and those that most often reached grade 3 or higher, are those for which the protocol required specific monitoring and management. No new safety concern was identified.

Crude incidence of grade 3 or higher events rises across the dose cohorts, from 50.0 per cent at 100 mg to 83.3 per cent at 400 mg. That ordering is not reproduced when the same events are expressed per unit of exposure, because follow-up differed between the cohorts; the exposure-adjusted rates are at Table 14.3.16.2 and the highest of them is at 200 mg. No statement about the effect of dose on the risk of a grade 3 or higher event is made from either measure.

13 DISCUSSION AND OVERALL CONCLUSIONS

Study KLV-7702 evaluated kelvaparib in advanced solid tumours with a homologous recombination repair defect. The maximum tolerated dose was 300 mg orally twice daily: dose-limiting toxicities occurred in 0 of 6, 2 of 7, 5 of 28 and 4 of 12 DLT-evaluable subjects at 100, 200, 300 and 400 mg, and 300 mg is the recommended phase 2 dose. Kelvaparib produced antitumor activity in advanced solid tumours with a homologous recombination repair defect (overall response rate 29.1 per cent); efficacy observations are exploratory. The safety profile was dominated by anaemia and myelosuppression, both on-target consequences of the mechanism of action. Myelodysplastic syndrome or acute myeloid leukaemia occurred in 3.4 per cent of subjects (Section 12.2.6). Limitations should be borne in mind. The study was open-label, which may influence investigator assessment of response and reporting of subjective adverse events. Follow-up was ongoing at the data cut-off and time-to-event estimates are therefore immature. The absence of a randomised comparator means no conclusion can be drawn about relative efficacy against available therapy. Comparison with historical data is not appropriate given differences in population and assessment criteria. Taken together, the results of study KLV-7702 support the continued development of kelvaparib at 300 mg, with weekly blood counts for the first month and monthly thereafter, in high-grade serous ovarian carcinoma.

The population enrolled was consistent with the intended population and with the eligibility criteria. Exposure was sufficient to characterise both the efficacy and the safety of the regimen. The analyses were performed as planned.

The population enrolled was the population the protocol described. Median age, performance status and prior therapy were consistent with the eligibility criteria, and there is no indication that the study enrolled a population easier to treat than the one intended.

Exposure was adequate. The median relative dose intensity indicates that the regimen as written could be delivered, and the pattern of reduction and interruption was consistent with the toxicities anticipated for oral PARP1-selective inhibitor.

The efficacy result should be read together with its confidence interval and with the maturity of the follow-up. The study had no concurrent control and the plan prespecifies no efficacy threshold, so the estimate is descriptive, and the contribution of the natural history of the disease cannot be separated from the contribution of treatment.

The safety profile was consistent with the mechanism and with the earlier clinical experience. The events reported most often, and those most often reaching grade 3 or higher, were those for which the protocol required specific monitoring, which indicates that the monitoring was directed at the right things.

10 subjects (17.2 per cent) discontinued study treatment because of an adverse event. That figure, rather than the overall incidence of events, is the measure of whether the regimen is tolerable in practice, because it counts the events that could not be managed by dose modification.

Deaths were reviewed individually. Those attributed to an adverse event are described in the narratives; the remainder are recorded as disease outcomes and are consistent with what would be expected in this population over this period.

The limitations of this study should be borne in mind. Follow-up at this analysis is limited and the longer time-to-event estimates are immature. The study was open-label, so investigator assessment of any subjective endpoint may have been influenced by knowledge of the assigned treatment. The subgroup analyses are descriptive.

Taken together, the results support the conclusions stated above and do not identify a safety concern that was not anticipated. Nothing in this analysis indicates that the assumptions underlying the design were materially wrong.

The limitations of the study should be borne in mind when interpreting these results. The absence of a randomised control means that the contribution of the natural history of the disease to the observed outcome cannot be quantified. Follow-up at the time of this analysis is limited, so the estimates of the longer time-to-event endpoints are immature and will change as further events accrue.

14 TABLES AND FIGURES REFERRED TO BUT NOT IN THE TEXT

The summary tables referred to in the text are presented in this section. Subject data listings are provided in Appendix 16.2.

14.1 Demographic data

Table 14.1.1 Demographic and Baseline Characteristics – Safety Analysis Set

Characteristic	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Age (years)					
Mean (SD)	65.8 (2.04)	66.2 (10.96)	66.4 (8.86)	66.6 (9.79)	66.3 (8.78)
Median	66.0	65.0	68.0	67.5	67.0
Min, max	63, 68	50, 86	50, 84	47, 80	47, 86
>= 65 years, n (%)	4 (66.7)	5 (55.6)	20 (64.5)	8 (66.7)	37 (63.8)
Sex, n (%)					
Male	3 (50.0)	3 (33.3)	21 (67.7)	10 (83.3)	37 (63.8)
Female	3 (50.0)	6 (66.7)	10 (32.3)	2 (16.7)	21 (36.2)
Race, n (%)					
White	6 (100.0)	5 (55.6)	26 (83.9)	7 (58.3)	44 (75.9)
Asian	0 (0.0)	1 (11.1)	1 (3.2)	3 (25.0)	5 (8.6)
Black or African American	0 (0.0)	2 (22.2)	3 (9.7)	1 (8.3)	6 (10.3)
Other	0 (0.0)	1 (11.1)	1 (3.2)	1 (8.3)	3 (5.2)
ECOG performance status, n (%)					
0	3 (50.0)	4 (44.4)	15 (48.4)	6 (50.0)	28 (48.3)
1	2 (33.3)	4 (44.4)	12 (38.7)	5 (41.7)	23 (39.7)
2	1 (16.7)	1 (11.1)	4 (12.9)	1 (8.3)	7 (12.1)
BRCA mutation status, n (%)					
BRCA1 mutated	0 (0.0)	2 (22.2)	10 (32.3)	3 (25.0)	15 (25.9)
BRCA2 mutated	3 (50.0)	0 (0.0)	7 (22.6)	2 (16.7)	12 (20.7)
BRCA wild-type	3 (50.0)	7 (77.8)	14 (45.2)	7 (58.3)	31 (53.4)
Homologous recombination deficiency, n (%)	3 (50.0)	4 (44.4)	17 (54.8)	8 (66.7)	32 (55.2)
Number of prior lines of therapy					
Median	1.5	1.0	2.0	1.0	1.0
>= 3, n (%)	2 (33.3)	0 (0.0)	6 (19.4)	1 (8.3)	9 (15.5)

SD = standard deviation; ECOG = Eastern Cooperative Oncology Group. Percentages are based on the number of subjects in the safety analysis set within each group.

Table 14.1.2 Subject Disposition – Safety Analysis Set

Disposition	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Subjects treated, n	6	9	31	12	58
Treatment ongoing at data cut-off, n (%)	3 (50.0)	2 (22.2)	8 (25.8)	4 (33.3)	17 (29.3)
Discontinued study treatment, n (%)	3 (50.0)	7 (77.8)	23 (74.2)	8 (66.7)	41 (70.7)
Disease progression	1 (16.7)	3 (33.3)	7 (22.6)	3 (25.0)	14 (24.1)
Adverse event	0 (0.0)	2 (22.2)	1 (3.2)	1 (8.3)	4 (6.9)
Death	1 (16.7)	0 (0.0)	4 (12.9)	1 (8.3)	6 (10.3)
Withdrawal by subject	1 (16.7)	1 (11.1)	5 (16.1)	1 (8.3)	8 (13.8)
Physician decision	0 (0.0)	1 (11.1)	2 (6.5)	2 (16.7)	5 (8.6)
Lost to follow-up	0 (0.0)	0 (0.0)	3 (9.7)	0 (0.0)	3 (5.2)
Protocol deviation	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)

Percentages are based on the number of subjects treated within each group.

Table 14.1.3 Extent of Exposure – Safety Analysis Set

Exposure parameter	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Duration of treatment (days)					
Median	171.0	117.0	118.0	147.0	121.5
Q1, Q3	107.8, 198.2	74.0, 129.0	83.5, 156.0	110.5, 226.2	86.0, 164.8
Min, max	65, 236	8, 308	8, 238	8, 448	8, 448
Cycles started, median	6.0	4.0	4.0	5.5	4.0
Relative dose intensity (%), median	83.8	83.9	84.7	82.9	84.2
Any dose reduction, n (%)	0 (0.0)	2 (22.2)	6 (19.4)	4 (33.3)	12 (20.7)
Any dose interruption, n (%)	1 (16.7)	3 (33.3)	5 (16.1)	1 (8.3)	10 (17.2)

Q1 = first quartile; Q3 = third quartile. Relative dose intensity is the total dose received divided by the total dose planned over the treatment period.

14.2 Efficacy data

Table 14.2.1 Best Overall Response by RECIST 1.1 – Full Analysis Set

Best overall response	100 mg (N=6)	200 mg (N=8)	300 mg (N=30)	400 mg (N=11)	Total (N=55)
Complete response (CR)	0 (0.0)	1 (12.5)	2 (6.7)	1 (9.1)	4 (7.3)
Partial response (PR)	0 (0.0)	2 (25.0)	6 (20.0)	4 (36.4)	12 (21.8)
Stable disease	4 (66.7)	1 (12.5)	13 (43.3)	3 (27.3)	21 (38.2)
Progressive disease	2 (33.3)	4 (50.0)	9 (30.0)	3 (27.3)	18 (32.7)
Not evaluable	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Objective response rate, n (%)	0 (0.0)	3 (37.5)	8 (26.7)	5 (45.5)	16 (29.1)
95% CI	(0.0, 45.9)	(8.5, 75.5)	(12.3, 45.9)	(16.7, 76.6)	(17.6, 42.9)

CI = confidence interval. Assessed according to RECIST version 1.1. Confidence intervals are exact (Clopper-Pearson).

Table 14.2.2 Kaplan-Meier Estimates of Time-to-event Endpoints – Full Analysis Set

Endpoint	100 mg (N=6)	200 mg (N=8)	300 mg (N=30)	400 mg (N=11)	Total (N=55)
Duration of response, median (months)	NE	NE	9.8	NE	NE
25th, 75th percentile	NE	NE	(9.8, NE)	(7.1, NE)	(9.8, NE)
Progression-free survival, median (months)	3.1	12.4	7.1	8.7	8.5
25th, 75th percentile	(2.9, 3.3)	(9.7, 12.7)	(3.0, 15.9)	(2.5, NE)	(3.0, 12.7)
Overall survival, median (months)	NE	NE	NE	NE	NE
25th, 75th percentile	NE	(12.7, NE)	NE	NE	NE

NE = not estimable. Duration of response is analysed among responding subjects only. It is measured from the first dose to the end of response recorded in the analysis dataset; response and progression dates cannot be re-derived because no assessment-level tumour data were collected in the database.

Table 14.2.3 Overall Response Rate and Progression-free Survival by BRCA Mutation Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
BRCA wild-type	28	9 (32.1)	(15.9, 52.4)	8.7
BRCA1 mutated	15	6 (40.0)	(16.3, 67.7)	5.8
BRCA2 mutated	12	1 (8.3)	(0.2, 38.5)	5.4

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.4 Overall Response Rate and Progression-free Survival by Homologous Recombination Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Negative	23	4 (17.4)	(5.0, 38.8)	8.5
Positive	32	12 (37.5)	(21.1, 56.3)	6.7

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.5 Overall Response Rate and Progression-free Survival by Prior Lines of Therapy – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
1 to 2	46	13 (28.3)	(16.0, 43.5)	8.7
>= 3	9	3 (33.3)	(7.5, 70.1)	4.8

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.6 Overall Response Rate and Progression-free Survival by Age Category – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
< 65 years	20	5 (25.0)	(8.7, 49.1)	8.9
>= 65 years	35	11 (31.4)	(16.9, 49.3)	7.1

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.7 Overall Response Rate and Progression-free Survival by Sex – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
Female	19	5 (26.3)	(9.1, 51.2)	9.7
Male	36	11 (30.6)	(16.3, 48.1)	5.9

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

Table 14.2.8 Overall Response Rate and Progression-free Survival by ECOG Performance Status – Full Analysis Set

Subgroup	n	Responders, n (%)	95% CI	Median PFS (months)
0	28	6 (21.4)	(8.3, 41.0)	5.9
1	21	10 (47.6)	(25.7, 70.2)	8.9
2	6	0 (0.0)	(0.0, 45.9)	3.5

CI = exact Clopper-Pearson confidence interval; NE = not estimable.

14.3 Safety data

Table 14.3.1 Overview of Treatment-emergent Adverse Events – Safety Analysis Set

Category, n (%)	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Any TEAE	6 (100.0)	9 (100.0)	31 (100.0)	12 (100.0)	58 (100.0)
Any treatment-related TEAE	5 (83.3)	9 (100.0)	31 (100.0)	11 (91.7)	56 (96.6)
Any grade >= 3 TEAE	3 (50.0)	7 (77.8)	21 (67.7)	10 (83.3)	41 (70.7)
Any serious TEAE	1 (16.7)	2 (22.2)	12 (38.7)	7 (58.3)	22 (37.9)
Any TEAE leading to dose reduction	0 (0.0)	2 (22.2)	6 (19.4)	4 (33.3)	12 (20.7)
Any TEAE leading to discontinuation	1 (16.7)	2 (22.2)	5 (16.1)	2 (16.7)	10 (17.2)
Any fatal TEAE	1 (16.7)	0 (0.0)	4 (12.9)	1 (8.3)	6 (10.3)

TEAE = treatment-emergent adverse event. A subject is counted once within each category.

Table 14.3.2 Treatment-emergent Adverse Events Occurring in at Least 2% of Subjects by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Gastrointestinal disorders	5 (83.3)	9 (100.0)	30 (96.8)	10 (83.3)	54 (93.1)
Nausea	2 (33.3)	5 (55.6)	21 (67.7)	9 (75.0)	37 (63.8)
Constipation	1 (16.7)	2 (22.2)	16 (51.6)	3 (25.0)	22 (37.9)
Vomiting	1 (16.7)	3 (33.3)	12 (38.7)	2 (16.7)	18 (31.0)
Abdominal pain	3 (50.0)	4 (44.4)	7 (22.6)	3 (25.0)	17 (29.3)
Diarrhoea	3 (50.0)	2 (22.2)	7 (22.6)	1 (8.3)	13 (22.4)
Dyspepsia	1 (16.7)	2 (22.2)	6 (19.4)	3 (25.0)	12 (20.7)
Blood and lymphatic system disorders	4 (66.7)	8 (88.9)	23 (74.2)	10 (83.3)	45 (77.6)
Anaemia	2 (33.3)	4 (44.4)	19 (61.3)	10 (83.3)	35 (60.3)
Neutropenia	2 (33.3)	4 (44.4)	8 (25.8)	3 (25.0)	17 (29.3)
Thrombocytopenia	1 (16.7)	2 (22.2)	10 (32.3)	3 (25.0)	16 (27.6)
Lymphopenia	1 (16.7)	2 (22.2)	4 (12.9)	3 (25.0)	10 (17.2)
Leukopenia	1 (16.7)	1 (11.1)	2 (6.5)	2 (16.7)	6 (10.3)
General disorders and administration site conditions	4 (66.7)	6 (66.7)	25 (80.6)	8 (66.7)	43 (74.1)
Fatigue	4 (66.7)	4 (44.4)	18 (58.1)	7 (58.3)	33 (56.9)
Asthenia	0 (0.0)	2 (22.2)	6 (19.4)	3 (25.0)	11 (19.0)
Pyrexia	1 (16.7)	0 (0.0)	7 (22.6)	1 (8.3)	9 (15.5)
Oedema peripheral	0 (0.0)	1 (11.1)	4 (12.9)	0 (0.0)	5 (8.6)
Nervous system disorders	2 (33.3)	3 (33.3)	16 (51.6)	5 (41.7)	26 (44.8)
Headache	0 (0.0)	1 (11.1)	10 (32.3)	4 (33.3)	15 (25.9)
Dizziness	2 (33.3)	1 (11.1)	6 (19.4)	0 (0.0)	9 (15.5)
Dysgeusia	1 (16.7)	1 (11.1)	4 (12.9)	1 (8.3)	7 (12.1)
Infections and infestations	2 (33.3)	4 (44.4)	14 (45.2)	5 (41.7)	25 (43.1)
Urinary tract infection	1 (16.7)	2 (22.2)	4 (12.9)	4 (33.3)	11 (19.0)
Nasopharyngitis	1 (16.7)	2 (22.2)	5 (16.1)	1 (8.3)	9 (15.5)
Pneumonia	0 (0.0)	1 (11.1)	3 (9.7)	1 (8.3)	5 (8.6)
Sepsis	0 (0.0)	0 (0.0)	2 (6.5)	0 (0.0)	2 (3.4)
Metabolism and nutrition disorders	2 (33.3)	5 (55.6)	12 (38.7)	5 (41.7)	24 (41.4)
Decreased appetite	2 (33.3)	5 (55.6)	12 (38.7)	5 (41.7)	24 (41.4)
Investigations	1 (16.7)	4 (44.4)	13 (41.9)	6 (50.0)	24 (41.4)
Alanine aminotransferase increased	0 (0.0)	1 (11.1)	6 (19.4)	2 (16.7)	9 (15.5)
Aspartate aminotransferase increased	1 (16.7)	0 (0.0)	4 (12.9)	3 (25.0)	8 (13.8)
Weight decreased	0 (0.0)	2 (22.2)	3 (9.7)	2 (16.7)	7 (12.1)
Blood creatinine increased	0 (0.0)	3 (33.3)	1 (3.2)	1 (8.3)	5 (8.6)

System organ class / Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Musculoskeletal and connective tissue disorders	0 (0.0)	4 (44.4)	13 (41.9)	3 (25.0)	20 (34.5)
Back pain	0 (0.0)	2 (22.2)	7 (22.6)	0 (0.0)	9 (15.5)
Arthralgia	0 (0.0)	1 (11.1)	5 (16.1)	2 (16.7)	8 (13.8)
Myalgia	0 (0.0)	2 (22.2)	2 (6.5)	1 (8.3)	5 (8.6)
Respiratory, thoracic and mediastinal disorders	0 (0.0)	1 (11.1)	8 (25.8)	3 (25.0)	12 (20.7)
Dyspnoea	0 (0.0)	1 (11.1)	3 (9.7)	2 (16.7)	6 (10.3)
Cough	0 (0.0)	0 (0.0)	5 (16.1)	1 (8.3)	6 (10.3)
Psychiatric disorders	2 (33.3)	1 (11.1)	7 (22.6)	1 (8.3)	11 (19.0)
Insomnia	2 (33.3)	1 (11.1)	7 (22.6)	1 (8.3)	11 (19.0)
Vascular disorders	0 (0.0)	1 (11.1)	7 (22.6)	0 (0.0)	8 (13.8)
Hypertension	0 (0.0)	1 (11.1)	7 (22.6)	0 (0.0)	8 (13.8)
Skin and subcutaneous tissue disorders	1 (16.7)	1 (11.1)	2 (6.5)	0 (0.0)	4 (6.9)
Rash	1 (16.7)	1 (11.1)	2 (6.5)	0 (0.0)	4 (6.9)
Neoplasms benign, malignant and unspecified	0 (0.0)	1 (11.1)	2 (6.5)	1 (8.3)	4 (6.9)
Myelodysplastic syndrome	0 (0.0)	1 (11.1)	1 (3.2)	0 (0.0)	2 (3.4)
Malignant neoplasm progression	0 (0.0)	0 (0.0)	1 (3.2)	1 (8.3)	2 (3.4)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 22.0.

Table 14.3.3 Treatment-emergent Adverse Events of Grade 3 or Higher by System Organ Class and Preferred Term – Safety Analysis Set

System organ class / Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Blood and lymphatic system disorders	2 (33.3)	4 (44.4)	16 (51.6)	9 (75.0)	31 (53.4)
Anaemia	1 (16.7)	1 (11.1)	10 (32.3)	6 (50.0)	18 (31.0)
Neutropenia	0 (0.0)	2 (22.2)	3 (9.7)	3 (25.0)	8 (13.8)
Thrombocytopenia	0 (0.0)	1 (11.1)	4 (12.9)	1 (8.3)	6 (10.3)
Leukopenia	1 (16.7)	0 (0.0)	2 (6.5)	0 (0.0)	3 (5.2)
Lymphopenia	0 (0.0)	0 (0.0)	2 (6.5)	1 (8.3)	3 (5.2)
Infections and infestations	0 (0.0)	1 (11.1)	5 (16.1)	0 (0.0)	6 (10.3)
Pneumonia	0 (0.0)	1 (11.1)	3 (9.7)	0 (0.0)	4 (6.9)
Sepsis	0 (0.0)	0 (0.0)	2 (6.5)	0 (0.0)	2 (3.4)
Gastrointestinal disorders	2 (33.3)	1 (11.1)	2 (6.5)	0 (0.0)	5 (8.6)
Ascites	1 (16.7)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.7)
Vomiting	1 (16.7)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.7)
Abdominal pain	0 (0.0)	1 (11.1)	0 (0.0)	0 (0.0)	1 (1.7)
Intestinal obstruction	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Nausea	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Neoplasms benign, malignant and unspecified	0 (0.0)	1 (11.1)	2 (6.5)	1 (8.3)	4 (6.9)
Myelodysplastic syndrome	0 (0.0)	1 (11.1)	1 (3.2)	0 (0.0)	2 (3.4)
Malignant neoplasm progression	0 (0.0)	0 (0.0)	1 (3.2)	1 (8.3)	2 (3.4)
Investigations	0 (0.0)	2 (22.2)	2 (6.5)	0 (0.0)	4 (6.9)
Blood creatinine increased	0 (0.0)	2 (22.2)	1 (3.2)	0 (0.0)	3 (5.2)
Aspartate aminotransferase increased	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Respiratory, thoracic and mediastinal disorders	0 (0.0)	0 (0.0)	2 (6.5)	0 (0.0)	2 (3.4)
Dyspnoea	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Cough	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Nervous system disorders	0 (0.0)	1 (11.1)	0 (0.0)	0 (0.0)	1 (1.7)
Headache	0 (0.0)	1 (11.1)	0 (0.0)	0 (0.0)	1 (1.7)

Subjects are counted once per preferred term and once per system organ class at their highest reported grade. MedDRA version 22.0.

Table 14.3.4 Treatment-related Treatment-emergent Adverse Events Occurring in at Least 2% of Subjects – Safety Analysis Set

Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Fatigue	3 (50.0)	2 (22.2)	12 (38.7)	4 (33.3)	21 (36.2)
Anaemia	0 (0.0)	2 (22.2)	13 (41.9)	6 (50.0)	21 (36.2)
Decreased appetite	1 (16.7)	5 (55.6)	8 (25.8)	2 (16.7)	16 (27.6)
Nausea	1 (16.7)	1 (11.1)	9 (29.0)	5 (41.7)	16 (27.6)
Constipation	0 (0.0)	1 (11.1)	10 (32.3)	2 (16.7)	13 (22.4)
Abdominal pain	3 (50.0)	3 (33.3)	5 (16.1)	1 (8.3)	12 (20.7)
Vomiting	0 (0.0)	3 (33.3)	7 (22.6)	1 (8.3)	11 (19.0)
Insomnia	2 (33.3)	1 (11.1)	6 (19.4)	1 (8.3)	10 (17.2)
Urinary tract infection	1 (16.7)	2 (22.2)	4 (12.9)	3 (25.0)	10 (17.2)
Neutropenia	0 (0.0)	1 (11.1)	5 (16.1)	3 (25.0)	9 (15.5)
Headache	0 (0.0)	1 (11.1)	6 (19.4)	1 (8.3)	8 (13.8)
Thrombocytopenia	0 (0.0)	2 (22.2)	4 (12.9)	2 (16.7)	8 (13.8)
Lymphopenia	1 (16.7)	1 (11.1)	3 (9.7)	1 (8.3)	6 (10.3)
Diarrhoea	1 (16.7)	1 (11.1)	3 (9.7)	1 (8.3)	6 (10.3)
Asthenia	0 (0.0)	1 (11.1)	3 (9.7)	2 (16.7)	6 (10.3)
Weight decreased	0 (0.0)	2 (22.2)	2 (6.5)	2 (16.7)	6 (10.3)
Dyspepsia	1 (16.7)	1 (11.1)	3 (9.7)	0 (0.0)	5 (8.6)
Leukopenia	1 (16.7)	1 (11.1)	1 (3.2)	2 (16.7)	5 (8.6)
Nasopharyngitis	0 (0.0)	2 (22.2)	2 (6.5)	1 (8.3)	5 (8.6)
Alanine aminotransferase increased	0 (0.0)	1 (11.1)	2 (6.5)	2 (16.7)	5 (8.6)
Arthralgia	0 (0.0)	1 (11.1)	2 (6.5)	2 (16.7)	5 (8.6)
Pyrexia	0 (0.0)	0 (0.0)	4 (12.9)	1 (8.3)	5 (8.6)
Dysgeusia	1 (16.7)	1 (11.1)	2 (6.5)	0 (0.0)	4 (6.9)
Dizziness	1 (16.7)	0 (0.0)	3 (9.7)	0 (0.0)	4 (6.9)
Back pain	0 (0.0)	1 (11.1)	3 (9.7)	0 (0.0)	4 (6.9)
Hypertension	0 (0.0)	1 (11.1)	3 (9.7)	0 (0.0)	4 (6.9)
Rash	1 (16.7)	1 (11.1)	1 (3.2)	0 (0.0)	3 (5.2)
Myalgia	0 (0.0)	1 (11.1)	1 (3.2)	1 (8.3)	3 (5.2)
Dyspnoea	0 (0.0)	1 (11.1)	1 (3.2)	1 (8.3)	3 (5.2)
Aspartate aminotransferase increased	1 (16.7)	0 (0.0)	1 (3.2)	0 (0.0)	2 (3.4)
Blood creatinine increased	0 (0.0)	1 (11.1)	0 (0.0)	1 (8.3)	2 (3.4)
Pneumonia	0 (0.0)	0 (0.0)	2 (6.5)	0 (0.0)	2 (3.4)
Oedema peripheral	0 (0.0)	0 (0.0)	2 (6.5)	0 (0.0)	2 (3.4)
Sepsis	0 (0.0)	0 (0.0)	2 (6.5)	0 (0.0)	2 (3.4)
Cough	0 (0.0)	0 (0.0)	1 (3.2)	1 (8.3)	2 (3.4)

Relationship assessed by the investigator. Percentages are based on the number of subjects treated in each group. Overall N = 58.

Table 14.3.5 Serious Treatment-emergent Adverse Events by Preferred Term – Safety Analysis Set

Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Anaemia	0 (0.0)	1 (11.1)	4 (12.9)	4 (33.3)	9 (15.5)
Blood creatinine increased	0 (0.0)	2 (22.2)	0 (0.0)	0 (0.0)	2 (3.4)
Malignant neoplasm progression	0 (0.0)	0 (0.0)	1 (3.2)	1 (8.3)	2 (3.4)
Pneumonia	0 (0.0)	0 (0.0)	2 (6.5)	0 (0.0)	2 (3.4)
Ascites	1 (16.7)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.7)
Aspartate aminotransferase increased	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Cough	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Intestinal obstruction	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Lymphopenia	0 (0.0)	0 (0.0)	0 (0.0)	1 (8.3)	1 (1.7)
Myelodysplastic syndrome	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Nausea	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Neutropenia	0 (0.0)	0 (0.0)	0 (0.0)	1 (8.3)	1 (1.7)
Sepsis	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Thrombocytopenia	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)

Subjects are counted once per preferred term. Percentages are based on N = 58.

Table 14.3.6 Treatment-emergent Adverse Events Leading to Discontinuation of Study Treatment – Safety Analysis Set

Preferred term	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Malignant neoplasm progression	0 (0.0)	0 (0.0)	1 (3.2)	1 (8.3)	2 (3.4)
Sepsis	0 (0.0)	0 (0.0)	2 (6.5)	0 (0.0)	2 (3.4)
Ascites	1 (16.7)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.7)
Intestinal obstruction	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Lymphopenia	0 (0.0)	0 (0.0)	0 (0.0)	1 (8.3)	1 (1.7)
Myelodysplastic syndrome	0 (0.0)	1 (11.1)	0 (0.0)	0 (0.0)	1 (1.7)
Pneumonia	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)
Thrombocytopenia	0 (0.0)	1 (11.1)	0 (0.0)	0 (0.0)	1 (1.7)

Percentages are based on N = 58.

Table 14.3.7 Deaths – Safety Analysis Set

Category, n (%)	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Any death	1 (16.7)	1 (11.1)	5 (16.1)	1 (8.3)	8 (13.8)
Death within 30 days of last dose	1 (16.7)	0 (0.0)	4 (12.9)	1 (8.3)	6 (10.3)
Primary cause of death					
Disease progression	0 (0.0)	1 (11.1)	0 (0.0)	0 (0.0)	1 (1.7)
Adverse event	1 (16.7)	0 (0.0)	4 (12.9)	1 (8.3)	6 (10.3)
Other	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	1 (1.7)

Percentages are based on the number of subjects treated within each group.

Table 14.3.8 Shift from Baseline to Worst Post-baseline CTCAE Grade for Selected Laboratory Parameters – Safety Analysis Set

Laboratory parameter	100 mg: worst grade 3 n (%)	100 mg: worst grade 4 n (%)	100 mg: shift >= 2 grades n (%)	200 mg: worst grade 3 n (%)	200 mg: worst grade 4 n (%)	200 mg: shift >= 2 grades n (%)	300 mg: worst grade 3 n (%)	300 mg: worst grade 4 n (%)	300 mg: shift >= 2 grades n (%)	400 mg: worst grade 3 n (%)	400 mg: worst grade 4 n (%)	400 mg: shift >= 2 grades n (%)	Total: worst grade 3 n (%)	Total: worst grade 4 n (%)	Total: shift >= 2 grades n (%)
Haemoglobin	0 (0.0)	0 (0.0)	2 (33.3)	1 (1.1)	2 (2.2)	4 (4.4)	10 (32.3)	4 (12.9)	17 (54.8)	3 (25.0)	0 (0.0)	5 (41.7)	14 (24.1)	6 (10.3)	28 (48.3)
Platelets	2 (33.3)	0 (0.0)	2 (33.3)	1 (1.1)	0 (0.0)	4 (4.4)	2 (6.5)	1 (3.2)	6 (19.4)	0 (0.0)	0 (0.0)	2 (16.7)	5 (8.6)	1 (1.7)	14 (24.1)
Neutrophils	0 (0.0)	0 (0.0)	2 (33.3)	1 (1.1)	1 (1.1)	3 (3.3)	4 (12.9)	0 (0.0)	4 (12.9)	1 (8.3)	0 (0.0)	1 (8.3)	6 (10.3)	1 (1.7)	10 (17.2)
Lymphocytes	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.1)	0 (0.0)	0 (0.0)	4 (12.9)	1 (3.2)	6 (19.4)	2 (16.7)	0 (0.0)	2 (16.7)	7 (12.1)	1 (1.7)	8 (13.8)
CA-125	2 (33.3)	0 (0.0)	1 (16.7)	0 (0.0)	0 (0.0)	2 (2.2)	4 (12.9)	2 (6.5)	10 (32.3)	2 (16.7)	0 (0.0)	4 (33.3)	8 (13.8)	2 (3.4)	17 (29.3)
ALT	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	5 (16.1)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.7)	0 (0.0)	5 (8.6)
AST	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.1)	1 (1.1)	2 (6.5)	0 (0.0)	3 (9.7)	2 (16.7)	1 (8.3)	3 (25.0)	4 (6.9)	2 (3.4)	7 (12.1)
Creatinine	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	3 (9.7)	0 (0.0)	3 (9.7)	1 (8.3)	0 (0.0)	2 (16.7)	4 (6.9)	0 (0.0)	5 (8.6)
Lactate dehydrogenase	1 (16.7)	0 (0.0)	1 (16.7)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (3.2)	6 (19.4)	1 (8.3)	1 (8.3)	2 (16.7)	2 (3.4)	2 (3.4)	9 (15.5)
Albumin	0 (0.0)	0 (0.0)	1 (16.7)	0 (0.0)	0 (0.0)	0 (0.0)	1 (3.2)	0 (0.0)	2 (6.5)	1 (8.3)	1 (8.3)	2 (16.7)	2 (3.4)	1 (1.7)	5 (8.6)

CTCAE = Common Terminology Criteria for Adverse Events, version 5.0. Percentages are based on the number of subjects treated in each group. Overall N = 58.

Table 14.3.9 Time to Onset and Resolution of the Most Frequent Treatment-emergent Adverse Events – Safety Analysis Set

Preferred term	Events, n	Median time to onset (days)	Resolved, n (%)	Median time to resolution (days)
Nausea	37	24.0	29 (78.4)	9.0
Anaemia	35	40.0	28 (80.0)	14.0
Fatigue	33	23.0	31 (93.9)	9.0
Decreased appetite	24	44.5	20 (83.3)	12.0
Constipation	22	39.0	19 (86.4)	8.0
Vomiting	18	23.0	14 (77.8)	5.5
Abdominal pain	17	46.0	15 (88.2)	10.0
Neutropenia	17	17.0	12 (70.6)	9.0
Thrombocytopenia	16	19.0	15 (93.8)	7.0
Headache	15	44.0	12 (80.0)	16.5
Diarrhoea	13	44.0	12 (92.3)	8.0
Dyspepsia	12	22.5	10 (83.3)	10.5
Insomnia	11	1.0	10 (90.9)	9.5
Urinary tract infection	11	51.0	8 (72.7)	15.5
Asthenia	11	30.0	8 (72.7)	8.5
Lymphopenia	10	38.0	10 (100.0)	17.0
Nasopharyngitis	9	8.0	6 (66.7)	12.0
Dizziness	9	30.0	6 (66.7)	9.5
Pyrexia	9	22.0	7 (77.8)	13.0
Alanine aminotransferase increased	9	43.0	8 (88.9)	6.5

Events are counted at the event level rather than the subject level. NE = not estimable.

14.3.10 Adverse events within subgroups

The summaries in Tables 14.3.10 to 14.3.15 are exploratory. The subgroups were not chosen or sized for comparisons of safety, the tables are not adjusted for multiplicity, and no conclusion about a difference between subgroups is drawn from them.

Table 14.3.10 Overview of adverse events by BRCA mutation status

Category	BRCA wild-type	BRCA1 mutated	BRCA2 mutated
Any event	31 (100.0)	15 (100.0)	12 (100.0)
Grade 3 or higher	25 (80.6)	10 (66.7)	6 (50.0)
Serious	11 (35.5)	6 (40.0)	5 (41.7)
Led to discontinuation	5 (16.1)	3 (20.0)	2 (16.7)
Fatal	1 (3.2)	3 (20.0)	2 (16.7)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.11 Overview of adverse events by homologous recombination status

Category	Negative	Positive
Any event	26 (100.0)	32 (100.0)
Grade 3 or higher	20 (76.9)	21 (65.6)
Serious	9 (34.6)	13 (40.6)
Led to discontinuation	4 (15.4)	6 (18.8)
Fatal	1 (3.8)	5 (15.6)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.12 Overview of adverse events by prior lines of therapy

Category	1 to 2	>= 3
Any event	49 (100.0)	9 (100.0)
Grade 3 or higher	37 (75.5)	4 (44.4)
Serious	21 (42.9)	1 (11.1)
Led to discontinuation	9 (18.4)	1 (11.1)
Fatal	5 (10.2)	1 (11.1)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.13 Overview of adverse events by age category

Category	<65	>=65
Any event	21 (100.0)	37 (100.0)
Grade 3 or higher	14 (66.7)	27 (73.0)
Serious	9 (42.9)	13 (35.1)
Led to discontinuation	5 (23.8)	5 (13.5)
Fatal	4 (19.0)	2 (5.4)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.14 Overview of adverse events by sex

Category	Female	Male
Any event	21 (100.0)	37 (100.0)
Grade 3 or higher	12 (57.1)	29 (78.4)
Serious	7 (33.3)	15 (40.5)
Led to discontinuation	6 (28.6)	4 (10.8)
Fatal	4 (19.0)	2 (5.4)

Percentages are calculated on the number of subjects in that subgroup.

Table 14.3.15 Overview of adverse events by ECOG performance status

Category	0	1	2
Any event	28 (100.0)	23 (100.0)	7 (100.0)
Grade 3 or higher	20 (71.4)	16 (69.6)	5 (71.4)
Serious	10 (35.7)	9 (39.1)	3 (42.9)
Led to discontinuation	5 (17.9)	3 (13.0)	2 (28.6)
Fatal	3 (10.7)	1 (4.3)	2 (28.6)

Percentages are calculated on the number of subjects in that subgroup.

14.4 Pharmacokinetic data

Table 14.4.1 Summary of Steady-state Pharmacokinetic Parameters by Dose Cohort – Pharmacokinetic Analysis Set

Dose cohort	n	Cmax (ng/mL) Geometric mean (%CV)	tmax (h) Median (min, max)	AUC0-12 (ng·h/mL) Geometric mean (%CV)	t1/2 (h) Mean (SD)	CL/F (L/h) Geometric mean (%CV)
100 mg	6	665 (34.2)	2.0 (2.0, 2.0)	6279 (44.8)	11.2 (5.07)	15.93 (44.8)
200 mg	9	1427 (29.3)	2.0 (2.0, 2.0)	14509 (34.3)	15.5 (7.61)	13.78 (34.3)
300 mg	31	1710 (31.4)	2.0 (2.0, 2.0)	17230 (37.5)	14.2 (5.34)	17.41 (37.5)
400 mg	12	2435 (32.2)	2.0 (2.0, 2.0)	25126 (34.5)	15.4 (4.87)	15.92 (34.5)

AUC0-12 = area under the concentration-time curve over one dosing interval; CL/F = apparent oral clearance; Cmax = maximum observed concentration; %CV = geometric coefficient of variation; t1/2 = terminal elimination half-life; tmax = time to maximum concentration.

Table 14.4.2 Accumulation Ratio and Trough Concentration by Dose Cohort – Pharmacokinetic Analysis Set

Dose cohort	n	Accumulation ratio Geometric mean (%CV)	Ctrough (ng/mL) Geometric mean (%CV)
100 mg	6	2.50 (23.6)	349.4 (67.3)
200 mg	9	2.96 (28.1)	916.3 (44.6)
300 mg	31	2.85 (21.5)	1070.7 (49.5)
400 mg	12	3.00 (18.9)	1626.0 (39.6)

Accumulation ratio is the ratio of AUC0-24 at steady state to AUC0-24 after the first dose.

15 REFERENCE LIST

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16 APPENDICES

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